

Troubleshooting Guide: Amine Sweetening Unit — Absorber Foaming and H2S Slip

Category: Gas Treating / Chemical Absorption **Unit:** Amine (MDEA/aMDEA) Sweetening Unit — Absorber Column **Tools:** HYSYS/UniSim Amine Property Package, absorber dP trending, foam tendency testing **Fluid Package:** Amine/Acid Gas package (Kent-Eisenberg or Li-Mather based electrolyte chemistry) — a generic PR/SRK cannot model chemical (reactive) absorption of CO₂/H₂S into amine **Symptom:** Sweet gas H₂S content drifting above pipeline spec despite normal lean amine circulation and concentration

1. Quick Reference

Item	Detail
Reported symptom	Sweet gas H ₂ S rising above the 4 ppm pipeline spec
Initially unclear	Amine quality issue vs. hydraulic/mass-transfer issue inside the absorber
Actual root cause	Absorber foaming caused by hydrocarbon condensate carryover and iron sulfide fines contaminating the amine, degrading tray/packing vapor-liquid contact
Fix	Improved inlet gas separation, added antifoam, installed amine filtration (particulate + carbon)
Diagnostic signal	Rising absorber differential pressure with normal lean amine concentration and circulation rate; foam confirmed at sample points
Prevention	Routine amine dP trending, scheduled foam-tendency testing, review of inlet separator performance

2. Symptom

- **Sweet gas H₂S content exceeded the 4 ppm pipeline specification.**
- **Lean amine circulation rate and concentration were both normal** — the parameters operators check first were not the cause.
- No equipment alarms had triggered yet, but gas quality was trending the wrong way.

3. Why This Wasn't an Obvious Amine-Quality Problem

H₂S slip is most often blamed on lean amine quality (low concentration, high heat-stable salts, degraded amine). Here, amine concentration and circulation checked out normal, which meant the absorption *chemistry* driving force was intact — so the investigation had to shift toward **mass transfer/hydraulics inside the column**, a different failure mode entirely (loss of effective vapor-liquid contact, not insufficient absorbent).

4. Diagnostic Approach

Step 1 — Confirm amine quality is not the driver

Lean amine concentration and circulation rate were checked against design and found normal, ruling out the most common cause.

Step 2 — Review absorber hydraulic trends

Absorber **differential pressure was rising** — a classic signature of reduced vapor space and increased liquid holdup on the trays, consistent with **foaming** rather than a chemistry deficiency.

Why the amine fluid package matters here: absorption of H₂S/CO₂ into amine is a **reactive** (chemical) absorption process, not simple physical VLE. Modeling or interpreting absorber performance correctly requires an amine-specific package (Kent-Eisenberg, Li-Mather, or equivalent) — a generic PR/SRK model would not correctly predict acid gas loading or column behavior for this system.

Step 3 — Confirm foaming directly

Foam was observed at sample/sight points on the column, and **foam tendency testing** on the amine sample confirmed a stable foam — physical evidence, not just an inference from dP trends.

Step 4 — Identify the foam-causing contaminant

Analysis of the amine identified **hydrocarbon condensate liquid carryover and iron sulfide (FeS) fines** as the foam stabilizers — both classic amine foaming culprits.

Quantitative Basis

- Sweet gas H₂S: baseline 2.1 ppm, rose to **9.8 ppm** against the 4 ppm pipeline spec.
- Absorber differential pressure: normal design range 0.30–0.45 bar; trended actual rose to **0.68 bar** — well above the design band despite unchanged gas/liquid loading.
- Lean amine held at **44.7 wt% MDEA** (design 45.0 wt%) and circulation held at design **850 gpm** throughout — both independently confirmed normal.
- Foam tendency test (ASTM D892-style): foam persisted **38 seconds** after air sparge stopped, versus a <10 second pass/fail criterion.
- Filtered-solids analysis: iron sulfide (FeS) content of **620 mg/L versus a <100 mg/L baseline**, confirming particulate-stabilized foam.

5. Root Cause

Hydrocarbon liquid carryover from the upstream inlet separator, combined with iron sulfide particulates, contaminated the amine and caused stable foaming in the absorber. The foam reduced effective vapor-liquid contact area and caused gas channeling, cutting mass transfer efficiency and allowing H₂S to slip into the sweet gas — even though the amine's underlying chemistry (concentration, circulation) remained healthy.

6. Corrective Action

1. Improved/verified inlet gas separator performance to reduce hydrocarbon carryover into the amine system.
2. Added antifoam to break the existing foam.

3. Installed particulate and activated-carbon filtration on the amine circuit to remove FeS fines and residual hydrocarbon.

7. Verification

- Absorber differential pressure returned to **0.38 bar**, within the 0.30–0.45 bar design band.
- Sweet gas H₂S content returned to **2.4 ppm**, within the 4 ppm spec, and held in the **2.0–2.8 ppm range over the following 30 days**.

8. Prevention / Long-Term Fix

- Established **routine absorber dP trending** as an early foaming indicator, independent of amine concentration checks.
 - Scheduled **periodic foam tendency testing** on the amine.
 - Reviewed inlet separator performance/maintenance to reduce the chance of repeat hydrocarbon carryover.
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9. General Troubleshooting Checklist (Reusable)

- ☐ Confirm lean amine concentration and circulation rate first (rules in/out the most common cause)
- ☐ If amine quality is normal, check absorber differential pressure trend for a foaming signature
- ☐ Physically confirm foam at sample/sight points before committing to an antifoam or filtration fix
- ☐ Run a foam tendency test on the amine sample
- ☐ Identify the specific contaminant (hydrocarbon carryover, FeS, degradation products, surfactants) rather than just treating with antifoam alone
- ☐ Trace contamination back to its source (inlet separator performance, corrosion products, upstream chemical injection)
- ☐ Add filtration appropriate to the contaminant identified
- ☐ Build absorber dP and foam-testing into routine monitoring, not just incident response

10. Key Takeaway

Amine quality (concentration, circulation) tells you about absorption **capacity**; absorber differential pressure tells you about absorption **hydraulics**. H₂S slip with healthy amine chemistry means the column isn't achieving proper vapor-liquid contact — check for foaming before assuming a chemistry problem, and always identify the specific contaminant driving the foam rather than treating symptoms with antifoam alone.

Related Concepts / Tags

amine-sweetening absorber-foaming h₂s-slip kent-eisenberg foam-tendency-test
hydrocarbon-carryover iron-sulfide gas-treating HYSYS UniSim

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.